Assignment 1: Natural Language Processing

Part (i): Trying a real plagiarism-detection tool

Tool used

Duplichecker (duplichecker.com), a free web-based plagiarism checker that compares submitted text against web sources.

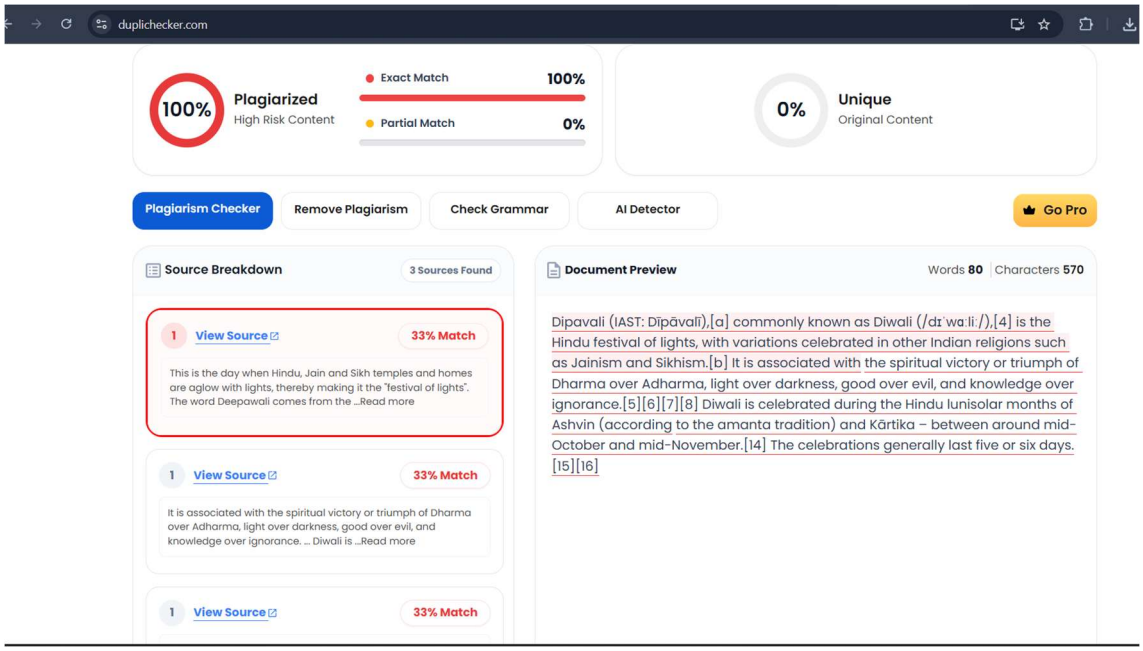
(a) Data given to the tool

Three text samples were pasted into the tool. All samples are about the festival of Diwali, so that Indian cultural content and an Indian language could be tested.

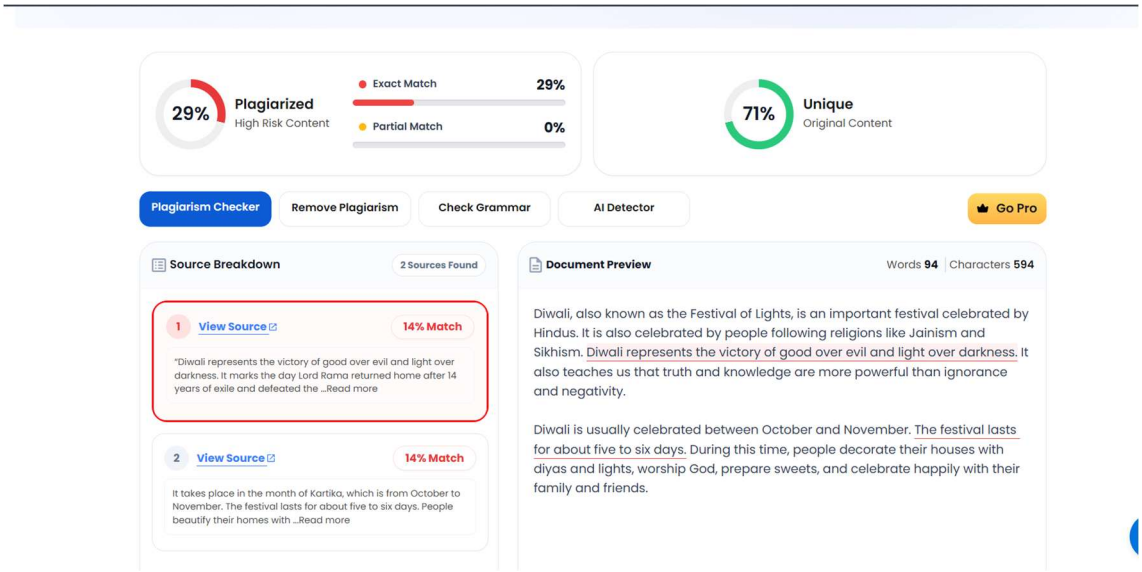
Sample	Content	Size	Purpose
Text 1	First paragraph of the English Wikipedia article on Diwali, copied directly	80 words, 570 characters	Exact copy in English
Text 2	The same ideas rewritten by me in my own words (paraphrase)	94 words, 594 characters	Paraphrase / semantic copying
Text 3	First paragraphs of the Marathi Wikipedia article on Diwali (Devanagari script), copied directly	104 words, 717 characters	Exact copy in an Indian language

Text 1 as pasted into the tool, shown in the Document Preview panel of the result (Screenshot 1), is the input that was checked.

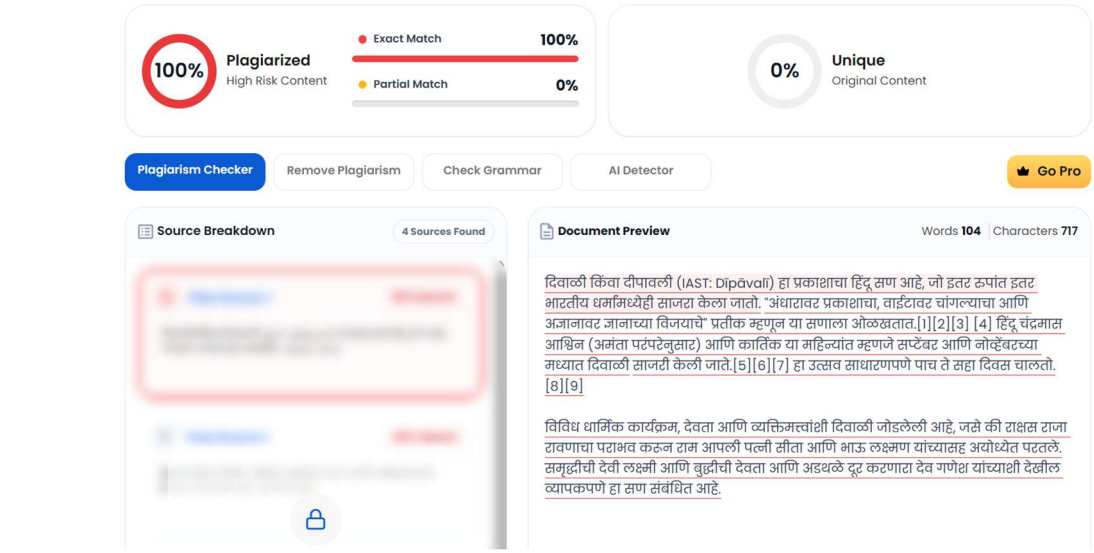
(b) How the system responded



Screenshot 1: Duplichecker result for Text 1 (English exact copy): 100% plagiarized, 3 sources found.



Screenshot 2: Duplichecker result for Text 2 (paraphrase): 29% plagiarized, 71% unique, 2 sources found.



Screenshot 3: Duplichecker result for Text 3 (Marathi exact copy): 100% plagiarized, 4 sources found, source details locked behind a Pro upgrade.

Sample	Plagiarized	Unique	Sources found	Observation
Text 1 (English copy)	100% (exact 100%, partial 0%)	0%	3 (33% match each)	Whole paragraph highlighted, including the Wikipedia citation numbers [5][6][7][8].
Text 2 (paraphrase)	29% (exact 29%, partial 0%)	71%	2 (14% match each)	Only two sentences highlighted: "Diwali represents the victory of good over evil and light over darkness" and "The festival lasts for about five to six days".
Text 3 (Marathi copy)	100% (exact 100%, partial 0%)	0%	4	Whole text highlighted. The source list was blurred with a lock icon, so the sources could not be viewed without paying.

(c) Where it succeeded and where it failed

Where it succeeded

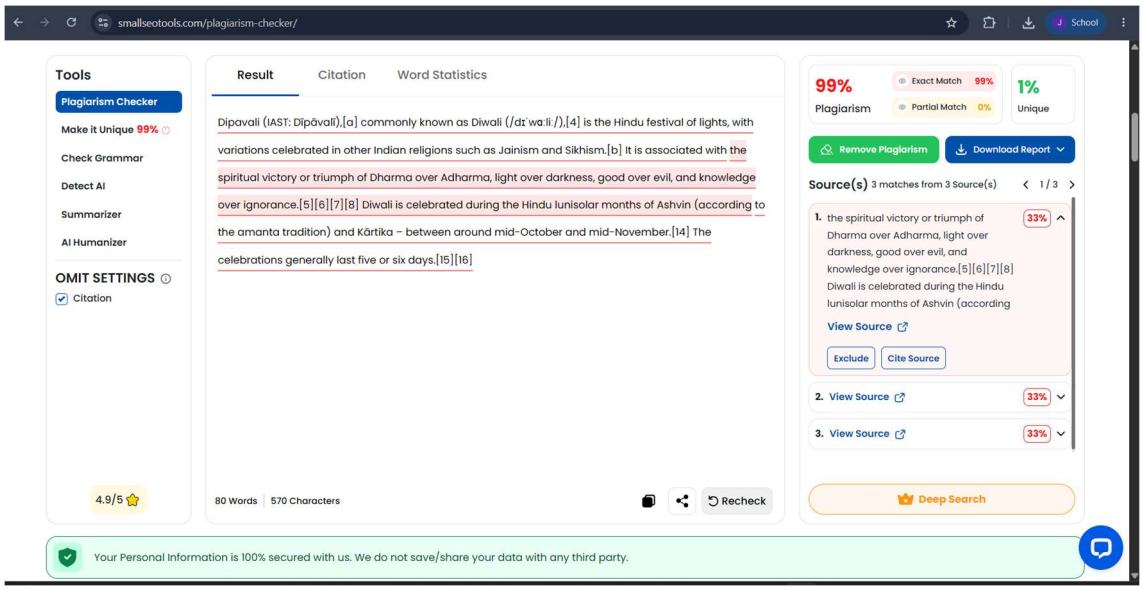
- The tool detected the English exact copy at 100% and listed the web pages the text came from.
- The tool detected the Marathi copy at 100%. This shows it can read Unicode Devanagari text and match it against Marathi web pages, not only English.
- The highlighting is shown at sentence level, so the user can see which parts were matched.

Where it failed or was weak

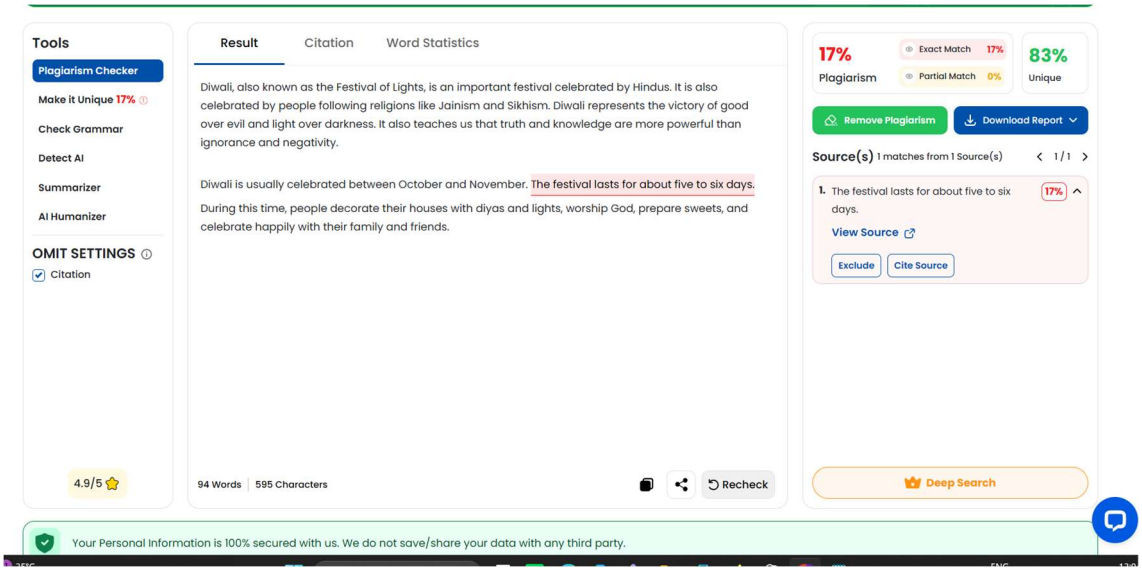
- Text 2 expresses the same facts (Hindu festival, good over evil, October to November, five to six days) in different words, yet the tool reported only 29%. Partial match was 0%, which means the tool does exact string matching and did not recognise meaning-level copying.
- It flagged the sentence about the victory of good over evil and light over darkness. This is a widely known idea about Diwali that anyone would write, so this is closer to a false positive on common cultural knowledge than real copying.
- For the Marathi text, the sources were hidden behind a paywall, so a student cannot verify what was matched.
- The tool counted Wikipedia citation markers such as [1][2][3] as part of the matched text, so formatting artefacts inflate the match.
- Translation or transliteration of Marathi text into English or Roman script was not tested here. Because the matching is string based, such a copy would be expected to be missed, and this is a limit of the approach.

Part (ii): Comparing two tools on the same input

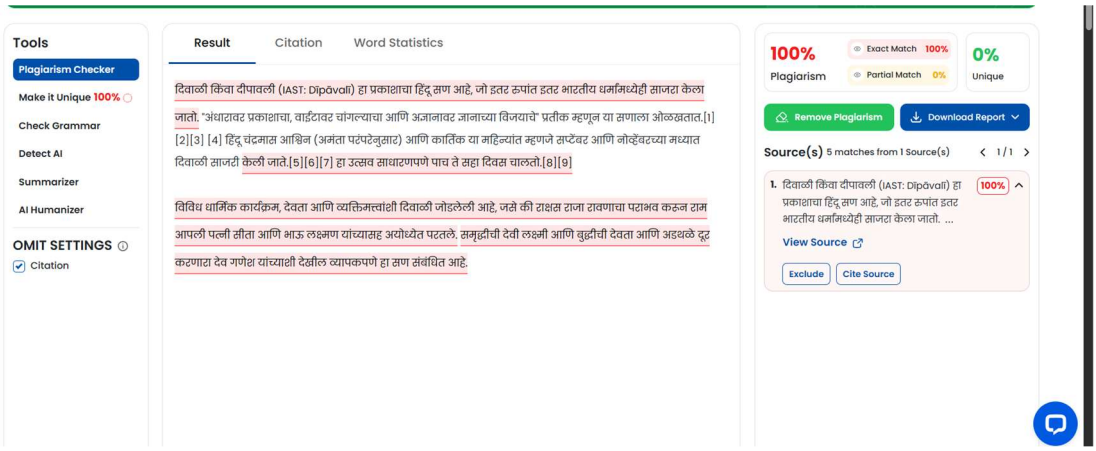
The same three samples were checked in a second tool, SmallSEOTools (smallseotools.com/plagiarism-checker). Duplichecker is Tool A and SmallSEOTools is Tool B.



Screenshot 4: SmallSEOTools result for Text 1 (English copy): 99% plagiarism, 3 sources.



Screenshot 5: SmallSEOTools result for Text 2 (paraphrase): 17% plagiarism, 83% unique, 1 source.



Screenshot 6: SmallSEOTools result for Text 3 (Marathi copy): 100% plagiarism, 5 matches from 1 source.

(a) Differences in outputs

Sample	Duplichecker (Tool A)	SmallSEOTools (Tool B)	Difference
Text 1: English copy	100% plagiarized, 0% unique, 3 sources at 33% each	99% plagiarism, 1% unique, 3 sources at 33% each	Almost identical. Tool B reports 1% as unique.
Text 2: Paraphrase	29% plagiarized, 71% unique, 2 sources at 14% each. Flagged 2 sentences.	17% plagiarism, 83% unique, 1 source at 17%. Flagged 1 sentence.	Tool A found more overlap. Tool B did not flag the good-over-evil sentence.
Text 3: Marathi copy	100% plagiarized, 4 sources, details locked	100% plagiarism, 5 matches from 1 source, matched source text visible	Same score. Tool B showed the matched Marathi source text, Tool A hid it behind Pro.

(b) Types of errors (error matrix)

Y = observed in the experiment, N = not observed, N/T = not tested.

Error type	Text 1 (English	Text 2 (paraphrase)	Text 3 (Marathi	Which tool

	copy)		copy)	
Semantic miss (paraphrase not detected)	N	Y (only 29% and 17% for fully copied ideas)	N	Both
False positive on common knowledge	N	Y ("victory of good over evil" flagged by Tool A)	N	Tool A
Context loss (citation markers counted as match)	Y ([5][6][7][8])	N	Y ([1][2][3], [8][9])	Both
Inconsistent score for same text	Y (100% vs 99%)	Y (29% vs 17%)	N (100% vs 100%)	A vs B
Source transparency problem	N	N	Y (sources locked)	Tool A
Grammatical error	N	N	N	None
Translation / cross-language miss	N/T	N/T	N/T	Not tested

(c) Cases of cultural or regional misunderstanding

- **Common cultural knowledge.** Both tools treat a general cultural statement, such as Diwali being the victory of good over evil and light over darkness, as if it were someone's property. Tool A flagged it even in my rewritten text. A system that only compares strings cannot tell shared cultural knowledge from copied writing.
- **Indian language handling.** Both tools matched the Marathi text only because it was copied letter for letter from a Marathi page. A paraphrase in Marathi, a Hindi version, or Marathi typed in Roman letters would not be matched in the same way, since they rely on matching the same words.
- **Source coverage.** The Marathi results look strong (100%), but Tool A locks the sources and Tool B showed only one source. Coverage of Marathi web content in these tools is therefore much smaller and less transparent than for English.

Part (iii): Adapting the tool for Marathi

Language chosen: Marathi (Devanagari script, spoken mainly in Maharashtra). Marathi is morphologically rich and has fewer digital resources than English, so a plagiarism checker needs more than a direct reuse of an English pipeline.

Concrete steps

1. **Collect a Marathi reference corpus.** Gather Marathi Wikipedia, news websites, books, theses and open-access journals, using public corpora such as IndicCorp and L3Cube-MahaCorpus as a start. This becomes the database that student text is compared with.
2. **Clean and normalise the text.** Convert to Unicode, fix different forms of the same character (for example nukta and conjunct variants), remove citation markers like [1][2], and standardise punctuation and digits (Devanagari and Western numerals).
3. **Tokenise and reduce words to a base form.** Use a Marathi tokenizer and sentence splitter (for example the Indic NLP Library), then apply a stemmer or lemmatizer. This is important because one Marathi word changes with case and gender suffixes (for example घर, घरात, घराचे), and exact matching would otherwise fail.

4. **Detect exact and near-exact copying.** Create word n-gram fingerprints (shingles) of each document and compare them with hashing methods such as MinHash or winnowing. Match on the stemmed forms, so that small changes in endings do not hide a copy.
5. **Detect paraphrased copying.** Convert sentences into vectors using a Marathi or multilingual language model (MahaBERT, MuRIL, IndicBERT or a multilingual sentence embedding) and compare by cosine similarity. This fixes the main weakness seen in Text 2.
6. **Detect cross-language copying.** Translate suspicious Marathi, Hindi or English text with a model such as IndicTrans2, or compare in a shared multilingual embedding space, so a Marathi text translated from English can still be found.
7. **Reduce false positives.** Keep a list of common phrases, proverbs, shlokas, abhangs and well-known facts, and lower the weight of matches on them. Ignore properly quoted and cited text.
8. **Evaluate and report.** Build a small test set of Marathi copies, paraphrases and original texts with human labels, measure precision and recall, and show the sources openly in the report, unlike the locked output seen in Tool A.

(a) Required linguistic resources

Resource	Why it is needed	Examples
Corpora	Reference database to compare against, and training data for models	Marathi Wikipedia, IndicCorp, L3Cube-MahaCorpus, Marathi news and thesis archives
Tokenizer and normaliser	Handle Devanagari Unicode, conjuncts, punctuation	Indic NLP Library
Stemmer / lemmatizer and morphological analyser	Match inflected word forms of the same word	Rule-based Marathi stemmers, Marathi morphological analysers
POS tagger	Identify content words and drop function words	Marathi POS taggers (for example from IIT Bombay or L3Cube MahaNLP)
Stop-word list	Remove words such as आणि, हा, ते that carry little meaning	Marathi stop-word lists
WordNet	Match synonyms in paraphrased text	Marathi WordNet (IIT Bombay)
Language models / embeddings	Measure meaning similarity for paraphrase detection	MahaBERT, MuRIL, IndicBERT, fastText Marathi vectors
Translation model and parallel corpora	Detect text translated from English or Hindi	IndicTrans2, Samanantar
Labelled evaluation set	Test accuracy on Marathi copies and paraphrases	To be created by annotators (no standard Marathi plagiarism dataset)

(b) Socio-linguistic challenges

- **Code-switching.** Marathi speakers often mix English words into Marathi sentences (for example "project submit kela"). The tool must handle both languages in one sentence, and a student can hide copying by replacing a few words with English ones.
- **Dialects.** Marathi varies by region (Varhadi, Khandeshi/Ahirani, Konkani-influenced and Marathwada speech). Words and endings differ, so a model trained mainly on standard written Marathi may miss dialect text.
- **Script variation.** Marathi can be written in Devanagari or typed in Roman letters ("Diwali ha prakashacha san aahe"). Roman spelling has no standard form, so matching needs transliteration into Devanagari first.

- **Spelling variation.** The same Marathi word can be spelled in more than one accepted way, which breaks exact matching unless text is normalised.
- **Low-resource language.** Marathi has much less digital text than English. Many sources are scanned PDFs or old books that are not searchable, so copying from them goes undetected.
- **Shared cultural content.** Proverbs, abhangs, shlokas, festival descriptions and oral stories are shared community heritage. The tool must not mark them as plagiarism, which means it needs a way to separate common cultural content from original authored text.
- **Related languages.** Marathi shares vocabulary with Sanskrit and Hindi, and Hindi text can be lightly edited into Marathi. This makes closely related language pairs a special case for the cross-language step.

Conclusion

Both tools detect exact copies in English and Marathi very well, but they are weak at finding paraphrased text and do not separate common cultural knowledge from copied writing. Adapting them for Marathi needs a Marathi corpus, proper normalisation and stemming, meaning-based comparison with language models, and handling of code-switching, dialects and script variation.